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# **Prediction of Ship Manoeuvring Hydrodynamic Coefficients Using Numerical Towing Tank Model Tests**

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Estimation of hydrodynamic derivatives or coefficients in the manoeuvring equations of motion is a very important and challenging task in the assessment of ship's manoeuvrability and in the prediction of its trajectory.

The relative demerits of experimental and theoretical methods paved way for numerical methods in the determination of hydrodynamic derivatives. The recent past has seen a host of numerical methods and they find easy implementation with the advance in the digital computing. FEM, BEM, strip theory etc. to name a few, often based on potential flow theory have been successfully used for the determination of first order hydrodynamic derivatives. However, these methods have failed in predicting all the derivatives due to lack of adequate flow physics and neglect of viscous effects.

With the emergence of RANSE based CFD as a powerful tool in flow prediction and associated phenomenon, many researchers have focused their interest in its applications to ship hydrodynamics such as resistance and propulsion, sea keeping and manoeuvring. Manoeuvring especially demands attention as safety issues are of prime concern as per IMO regulations. RANSE codes are being utilized to numerically simulate hydrodynamic experiments with an intention to reduce, and ultimately to replace, the physical model tests. The recent contributions in manoeuvring from CFD include calculations of forces acting on the ship during manoeuvring motion (Nonaka et-al, 2007), simulations of manoeuvres such as turning circle and zig-zag with rudder and propeller also modeled (Xing-Kaeding and Jensen, 2006), determination of linear hydrodynamic derivatives of a passenger ship undergoing forced motions (Cura-Hochbaum, 2006) and determination of forces on the ship hull using finite volume simulations of straight line test, circular motion test and planar motion mechanism tests (Ohmori, 1998). It can be understood that the method is not full-fledged to totally replace experiments. Yet it definitely promises to contribute a lot to early design calculations.

This paper brings about the detailed mathematical formulation and numerical procedure for simulating dynamic manoeuvres of a container ship model. The prescribed body motions have been brought about using user defined functions (UDF) such that the conventional horizontal planar motion mechanism (HPMM) test is duplicated. Non-linear mathematical model proposed by Son and Nomoto (Fossen, 1994) has been used. The computations of flow coupled with rigid body motions have been carried out using a RANSE based CFD solver employing finite volume technique for solving viscous flow equations. All the twenty one hydrodynamic derivatives appearing in the mathematical model have been determined and have been compared with published experimental results and are found to be satisfactory except in the case of two or three coefficients.

Any manoeuvring problem involving estimation of hydrodynamic derivatives starts with the selection of an appropriate mathematical model. In this work the model proposed by Son and Nomoto, 1982 (Fossen, 1994) has been used after neglecting the roll effects and the simplified equations of speed and steering in their non-dimensional form are given by Equations (1) to (3).

$$X' = X'(u)' + (1-t)T' + X'_{vr} v'r' + X'_{vv} v'^2 + X'_{rr} r'^2 + X'_\delta \sin \delta + X'_{ext} \quad (1)$$

$$Y' = Y'_v v' + Y'_r r' + Y'_{vvv} v'^3 + Y'_{rrr} r'^3 + Y'_{vvr} v'^2 r' + Y'_{vrr} v' r'^2 + Y'_\delta \cos \delta + Y'_{ext} \quad (2)$$

$$N' = N'_v v' + N'_r r' + N'_{vvv} v'^3 + N'_{rrr} r'^3 + N'_{vvr} v'^2 r' + N'_{vrr} v' r'^2 + N'_\delta \cos \delta + N'_{ext} \quad (3)$$

Where  $X'(u)$  is velocity dependent damping function, for instance  $X'(u) = X'_{|u|u}|u|$ . This model does not involve modeling the rudder and propeller. Hull derivatives are of prime interest and the effect of rudder (terms associated with  $\delta$ ) and propeller ( terms associated with  $t$  ) can be brought in by some empirical relationships when required. Excitation forces from environment (terms associated with  $ext$ ) are neglected.

The formulation of the problem has in fact been tedious, as it involved a laborious procedure of many trials to arrive at the expressions for body oscillations so that they in turn yield the most simplified expressions for the calculation of hydrodynamic derivatives. Here the Fourier Series Expansion method has been used for the determination of the derivatives according to which the hydrodynamic forces and moments on the ship hull can be expressed as Hydrodynamic

$$\text{force/moment} = \sum_{m=0}^3 (a_m \cos m\omega t + b_m \sin m\omega t)$$

$$\text{where the Fourier constants are given by } a_m = \frac{2}{T} \int_0^T f(t) \cos m\omega t \text{ and } b_m = \frac{2}{T} \int_0^T f(t) \sin m\omega t$$

Fourier series expansion has been considered only up to third harmonic as the mathematical model contains only up to third order terms.  $u, \dot{u}, v, \dot{v}, r$  and  $\dot{r}$  in Equations (1) to(3) have been substituted with sinusoidal expressions while  $X', Y'$  and  $N'$  have been expanded in Fourier series as explained above in three separate modes of motion viz. pure sway, pure yaw and combined sway and yaw (mixed mode). Comparison of like terms in the corresponding equations has yielded the following expressions for the hydrodynamic derivatives;

$$X_{vv} = \frac{2}{y_{0a}^2 \omega^2} a_{xy2} \quad (4)$$

$$X_{u|u|} = \frac{1}{V^2} [a_{xy0} - a_{xy2}] \quad (5)$$

$$Y_{\dot{v}} = \frac{1}{y_{0a} \omega^2} b_{yY1} \quad (6)$$

$$Y_{vvv} = \frac{-4}{y_{0a}^3 \omega^3} a_{yY3} \quad (7)$$

$$Y_v = \frac{1}{y_{0a} \omega} [3a_{yY3} - a_{yY1}] \quad (8)$$

$$N_{\dot{v}} = \frac{1}{y_{0a} \omega^2} b_{nN1} \quad (9)$$

$$N_{vvv} = \frac{-4}{y_{0a}^3 \omega^3} a_{nY3} \quad (10)$$

$$N_v = \frac{1}{y_{0a} \omega} [3a_{nY3} - a_{nY1}] \quad (11)$$

$$X_{\dot{u}} = \frac{-1}{2u_{aN}} b_{xN2} \quad (12)$$

$$X_{rr} = \frac{2}{r_a^2} \left[ \frac{2u_{cN}u_{aN}}{V^2} (a_{xY0} - a_{xY2}) - a_{xN2} \right] \quad (13)$$

$$Y_{\dot{r}} = \frac{1}{r_a \omega} a_{yN1} \quad (14)$$

$$Y_{rrr} = \frac{-4}{r_a^3} \left[ \frac{b_{xN2}r_a}{4\omega} + b_{yN3} \right] \quad (15)$$

$$Y_r = \frac{1}{r_a} \left[ b_{yN1} + \frac{b_{xN2}r_a}{2u_{aN}\omega} (u_{cN} - \frac{u_{aN}}{2}) + 3 \left( \frac{b_{xN2}r_a}{4\omega} + b_{yN3} \right) \right] \quad (16)$$

$$N_{\dot{r}} = \frac{1}{r_a \omega} a_{nN1} \quad (17)$$

$$N_{rrr} = \frac{-4}{r_a^3} b_{nN3} \quad (18)$$

$$N_r = \frac{1}{r_a} [b_{nN1} + 3b_{nN3}] \quad (19)$$

$$X_{vr} = \frac{2}{v_{aYN}r_a} b_{xYN2} - \frac{b_{yY1}}{y_{0a}\omega^2} - b_{xN2} \frac{u_{aYN}}{u_{aN}} \quad (20)$$

$$Y_{vvr} = \frac{4}{v_{aYN}^2 r_a} \left[ b_{yYN3} - \frac{b_{xN2}r_0}{4\omega} \left( \frac{u_{aYN}}{u_{aN}} + 1 \right) - b_{yN3} \right] \quad (21)$$

$$Y_{vrr} = \frac{-4}{v_{aYN}r_a^2} \left[ a_{yYN3} + a_{yY3} \left( \frac{v_{aYN}}{v_a} \right)^3 \right] \quad (22)$$

$$N_{vvr} = \frac{4}{v_{aYN}^2 r_a} [b_{nYN3} - b_{nN3}] \quad (23)$$

$$N_{vrr} = \frac{-4}{v_{aYN}r_a^2} \left[ a_{nYN3} + a_{nY3} \left( \frac{v_{aYN}}{v_a} \right)^3 \right] \quad (24)$$

In the above equations subscript ‘a’ indicates amplitude of motion. Subscripts ‘Y’, ‘N’ and ‘YN’ indicate pure sway, pure yaw and mixed modes respectively.

Simulation of HPMM requires a numerical computational domain and a solver capable of solving the fluid flow equations while the mesh in the fluid domain re-orient with the hull motion. ICEM CFD has been used for mesh generation and the dynamic mesh motion facility of the commercial CFD package FLUENT has been made use of to bring about the prescribed motion of the ship body.

The size of the virtual towing tank used in the CFD modeling has been fixed as per ITTC guidelines (ITTC quality manual, 1999) where the domain extends one ship length each from the port and starboard, 0.8 ship length from the fore, 1.5 ship lengths from the aft and 0.64 ship length downwards from the hull bottom. A 3-D unstructured grid with approximately 300,000 cells has been used for the analysis. The block structured grid system has not been successful in producing oscillations in one complete cycle due to the creation of negative volumes. Remeshing is required as the mesh on the ship surface moves and this can be easily brought about by the solver in case of an unstructured grid system than its structured counterpart. Meshing has been

carried out in a number of trials such that it is neither too coarse nor too fine and also uniformly graded through out the fluid domain.

A user defined function (UDF) has been written in C language using the macro 'DEFINE\_CG\_MOTION' and has been hooked into the main program in order to oscillate the mesh on the hull surface. So, the solution process also consists of updating the mesh position at each time step (0.005 sec, 20 iterations per time step). Free-surface effects have been neglected in the present study. Flow velocity has been taken 1 m/sec and time period of oscillation 13.33 sec. Values for motion velocity amplitudes were chosen such that they are neither too high nor too low. Many trials have been carried out with various velocity amplitudes and the most feasible ones have been chosen. Higher amplitudes caused mesh distortion and consequently dynamic mesh failure whereas lower amplitudes have not been able to capture the non-linear part of the forces and moments. The expressions for hydrodynamic derivatives given by Equations (4) to (24) have been expressed in terms of Fourier constants that have been obtained by the numerical integration of the time histories of forces and moments obtained through the simulation over a cycle using Simpson's rule. Table 1 shows the values of hydrodynamic derivatives calculated through the method adopted in this paper and their comparison with published experimental values (Fossen, 1994). The derivatives have also been assigned symbols depending on their relative importance (Strom-Tejsen, 1965) in trajectory prediction and stability characteristics.

The numerical simulation of HPMM using RANSE based CFD methods has been found to work well at least for moderate motion amplitudes. Fourier series expansion method used here for arriving at the expressions for derivatives is indeed a promising one and has been successful enough in giving all the 21 derivatives-coupled, uncoupled and cross-coupled targeted at in the mathematical model. Most values have been predicted well and higher deviations are seen in the values of  $Y'_r$  and  $N'_{vv}$ . These deviations may be due to the coarseness of the mesh or errors in numerical integration. As these derivatives fall in the most important category, they have to be predicted more accurately. The research is still on for obtaining better values of derivatives. This work on the whole finds its success in exploring and exploiting various features of a general purpose fluid flow solver to solve a non-linear dynamic ship manoeuvring problem.

**Table 1 Hydrodynamic Derivatives**

| Hydrodynamic Derivative | Non-Dimensionalisation factor | Type          | Present CFD | Experiment (Fossen, 1994) | Percentage Deviation | Category (Strom-Tejsen, 1965) |
|-------------------------|-------------------------------|---------------|-------------|---------------------------|----------------------|-------------------------------|
| $X'_{\dot{u}}$          | $0.5\rho L^3$                 | uncoupled     | -0.00027    | -0.00024                  | 13.44                | A                             |
| $X'_{u u }$             | $0.5\rho L^2$                 | „             | -0.00046    | -0.00042                  | 7.66                 | A                             |
| $X'_{vv}$               | $0.5\rho L^2$                 | coupled       | -0.00522    | -0.00386                  | 35.23                | C                             |
| $X'_{rr}$               | $0.5\rho L^4$                 | „             | 0.000209    | 0.0002                    | 4.5                  | C                             |
| $X'_{vr}$               | $0.5\rho L^3$                 | cross-coupled | -0.00297    | -0.00311                  | -4.47                | D                             |
| $Y'_{\dot{v}}$          | $0.5\rho L^3$                 | uncoupled     | -0.0069     | -0.00705                  | -2.11                | A                             |
| $Y'_{\dot{v}}$          | $0.5\rho L^2 V$               | „             | -0.01065    | -0.0116                   | -8.18                | A                             |
| $Y'_{vvv}$              | $0.5\rho L^2 / V$             | „             | -0.119      | -0.109                    | 9.17                 | C                             |
| $Y'_{\dot{r}}$          | $0.5\rho L^4$                 | coupled       | -0.00058    | -0.00035                  | 64.51                | A                             |
| $Y'_{\dot{r}}$          | $0.5\rho L^3 V$               | coupled       | 0.00212     | 0.00242                   | -12.39               | A                             |
| $Y'_{rrr}$              | $0.5\rho L^5 / V$             | coupled       | 0.00146     | 0.00177                   | 17.51                | D                             |
| $Y'_{vvr}$              | $0.5\rho L^3 / V$             | cross-coupled | 0.0179      | 0.0214                    | -16.35               | A                             |
| $Y'_{vrr}$              | $0.5\rho L^4 / V$             | „             | -0.0474     | -0.04605                  | 2.93                 | D                             |
| $N'_{\dot{v}}$          | $0.5\rho L^4$                 | coupled       | -0.00037    | -0.00035                  | 5.53                 | A                             |
| $N'_{\dot{v}}$          | $0.5\rho L^3 V$               | „             | -0.00429    | -0.00385                  | 11.29                | A                             |
| $N'_{vvv}$              | $0.5\rho L^3 / V$             | „             | 0.001154    | -0.001492                 | -22.65               | C                             |
| $N'_{\dot{r}}$          | $0.5\rho L^5$                 | uncoupled     | -0.00044    | -0.00042                  | 4.77                 | A                             |
| $N'_{\dot{r}}$          | $0.5\rho L^4 V$               | „             | -0.00211    | -0.00222                  | -5.0                 | A                             |
| $N'_{rrr}$              | $0.5\rho L^6 / V$             | „             | -0.0022     | -0.00229                  | -3.97                | D                             |
| $N'_{vvr}$              | $0.5\rho L^4 / V$             | cross-coupled | -0.0164     | -0.0424                   | -61.32               | A                             |
| $N'_{vrr}$              | $0.5\rho L^5 / V$             | cross-coupled | 0.0011      | 0.00156                   | -29.48               | D                             |

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## Nomenclature

|              |                                                                                    |
|--------------|------------------------------------------------------------------------------------|
| $a_m$        | Fourier constants associated with cosine terms                                     |
| $a_{nNm}$    | Fourier constants associated with cosine terms of yaw equation in pure yaw mode    |
| $a_{nYm}$    | Fourier constants associated with cosine terms of yaw equation in pure sway mode   |
| $a_{nYNm}$   | Fourier constants associated with cosine terms of yaw equation in mixed mode       |
| $a_{xNm}$    | Fourier constants associated with cosine terms of surge equation in pure yaw mode  |
| $a_{xYm}$    | Fourier constants associated with cosine terms of surge equation in pure sway mode |
| $a_{xYNm}$   | Fourier constants associated with cosine terms of surge equation in mixed mode     |
| $a_{yNm}$    | Fourier constants associated with cosine terms of sway equation in pure yaw mode   |
| $a_{yYm}$    | Fourier constants associated with cosine terms of sway equation in pure sway mode  |
| $a_{yYNm}$   | Fourier constants associated with cosine terms of sway equation in mixed mode      |
| $A$          | Derivatives evaluated and deemed very important                                    |
| $b_m$        | Fourier constants associated with sine terms                                       |
| $b_{nNm}$    | Fourier constants associated with sine terms of yaw equation in pure yaw mode      |
| $b_{nYm}$    | Fourier constants associated with sine terms of yaw equation in pure sway mode     |
| $b_{nYNm}$   | Fourier constants associated with sine terms of yaw equation in mixed mode         |
| $b_{xNm}$    | Fourier constants associated with sine terms of surge equation in pure yaw mode    |
| $b_{xYm}$    | Fourier constants associated with sine terms of surge equation in pure sway mode   |
| $b_{xYNm}$   | Fourier constants associated with sine terms of surge equation in mixed mode       |
| $b_{yNm}$    | Fourier constants associated with sine terms of sway equation in pure yaw mode     |
| $b_{yYm}$    | Fourier constants associated with sine terms of sway equation in pure sway mode    |
| $b_{yYNm}$   | Fourier constants associated with sine terms of sway equation in mixed mode        |
| $C$          | Derivatives evaluated and deemed of minor importance                               |
| $D$          | Derivatives evaluated and deemed negligible                                        |
| $L$          | Length between perpendiculars of the vessel                                        |
| $m$          | Integer for determining harmonics of Fourier series                                |
| $N'$         | Non-dimensional hydrodynamic reaction moment in yaw                                |
| $r, \dot{r}$ | Yaw rate, yaw acceleration of the model                                            |
| $T$          | Period of oscillation                                                              |
| $u, \dot{u}$ | Surge velocity, surge acceleration of the model                                    |
| $v, \dot{v}$ | Sway velocity, sway acceleration of the model                                      |
| $V$          | Forward velocity of the vessel                                                     |
| $X'$         | Non-dimensional hydrodynamic reaction force in surge                               |
| $Y'$         | Non-dimensional hydrodynamic reaction force in sway                                |
| $N'$         | Non-dimensional hydrodynamic moment in yaw                                         |
| $\gamma$     | Arbitrary value of phase which depends on the start value of oscillation           |
| $\omega$     | Circular frequency of oscillation                                                  |